خدمتكم طريق خضناه لرضى الله

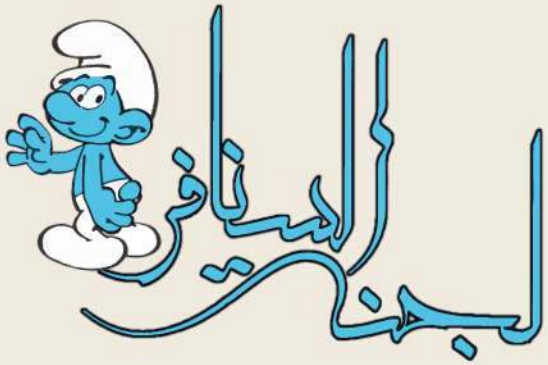
2021

أسئلة سنوات

ميد

مختبر فيزياء 1





بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق

(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتمل الصواب وقد تحتمل الخطأ

فإن أصبنا فما هو إلا توفيق من الله

وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

#لجنة_السنافر

#هي_لله

Question 9

Correct

Mark 1.00 out of 1.00

Flag question

The micrometer is more accurate than the vernier caliper

Select one:

a. False

b. True ✓

The correct answer is: True

Question 10

Correct

Mark 1.00 out of 1.00

Flag question

The using of the air track in Newtons second law experiment is to decrease the friction effects

Select one:

- a. True ✓
- b. False

The correct answer is: True

Question 11

Correct

Mark 1.00 out of 1.00

Flag question

Round the number 62.5347 to four significant figures

Select one:

- a. 62.50
- b. 63.00
- c. 62.53 ✓
- d. 62.54
- e. Non of above

The correct answer is: 62.53

Question 12

Correct

Mark 1.00 out of 1.00

Flag question

In the elastic collisions, the linear momentum is conserved, while the kinetic energy is not:

Select one:

a. True

b. False ✓

The correct answer is: False

Question 13

Incorrect

Mark 0.00 out of 1.00

Flag question

If F is the force, x the displacement, and k a spring constant, for simple harmonic motion, F is given by:

Select one:

a. F=kx

b. F = -kx

c. F=-k/x ✗

d. $F = -kx^2$

e. Non of above

The correct answer is: F = -kx

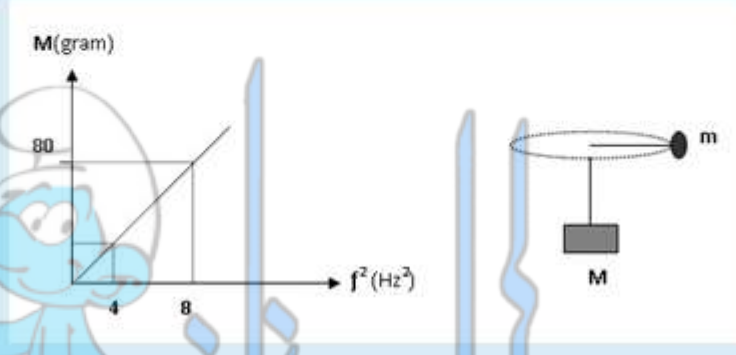
Question 14

Correct

Mark 1.00 out of 1.00

Flag question

In the circular motion experiment, a student studied the relation between the hanging mass (M) and the square of the frequency (f^2) for rotational body (m) using the radius (R=0.3 m). A plotted data are shown in the figure bellow. The mass (m) of the rotational body is:



Select one:

- a. 6.34 gram
- b. 8.44 gram ✓
- c. 12.7 gram
- d. Non of above
- e. 4.22 gram

The correct answer is: 8.44 gram

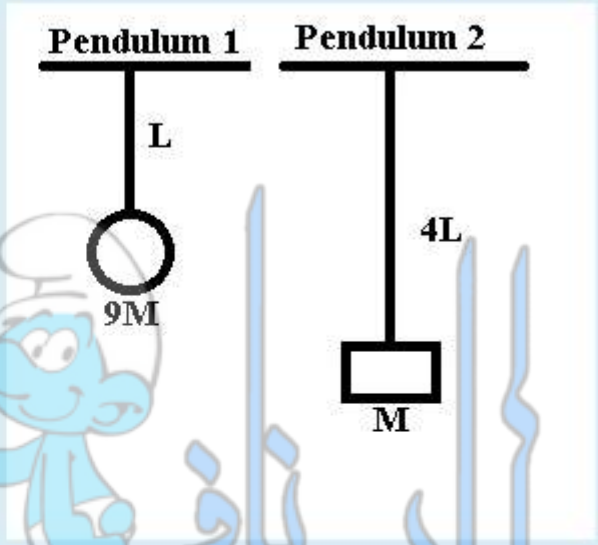
Question 15

Correct

Mark 1.00 out of 1.00

Flag question

In the simple pendulum experiment, the following figures show two pendulums. If the periodic time is T_1 for the Pendulum (1), and the periodic time is T_2 for the pendulum (2), then the relation between T_1 and T_2 is



Select one:

a. T2 = 2.45T1

b. Non of above

c. T2 = 2.24T1

d. T2=2T1 ✓

e. T2=1.73T1

The correct answer is: T2=2T1

Question 16

Correct

Mark 1.00 out of 1.00

Flag question

The best equipment to measure the thickness of the paper is the Vernier caliper

Select one:

a. True

b. False ✓

The correct answer is: False

Question 17

Correct

Mark 1.00 out of 1.00

Flag question

The sum of the change in kinetic and potential energy is always positive

Select one:

a. True

b. False ✓

The correct answer is: False

Question 1

Correct

Mark 1.00 out of 1.00

Flag question

In the simple harmonic oscillator, a mass of (40g) is hanged in a spring, if the extension of the spring is (4 cm) and if the mass drawn to (3cm) from equilibrium point and oscillated 25 times, the total time of an oscillation of 20 times is (in sec):

Select one:

- a. 9.93
- b. 11.91
- c. Non of above
- d. 5.96
- e. 7.95 ✓

The correct answer is: 7.95

Question 2

Correct

Mark 1.00 out of 1.00

Flag question

In vector experiment, the mass hangers do not affect on the equilibrium state

Select one:

a. False ✓

b. True

The correct answer is: False

Question 3

Correct

Mark 1.00 out of 1.00

Flag question

In uniform circular motion, the peed is constant but the velocity is variable

Select one:

a. True

b. False

The correct answer is: True

Question 4

Correct

Mark 1.00 out of 1.00

Flag question

In spring constant experiment, if the hanging mass 80g and the elongation in the spring is 4cm, the spring constant (N/m) is:

Select one:

- a. 40
- b. 20 ✓
- c. 10
- d. Non of above
- e. 5

The correct answer is: 20

Question 5

Correct

Mark 1.00 out of 1.00

Flag question

We can conclude from the frictional force that μ_k is smaller than μ_s

Select one:

- a. True ✓
- b. False

The correct answer is: True

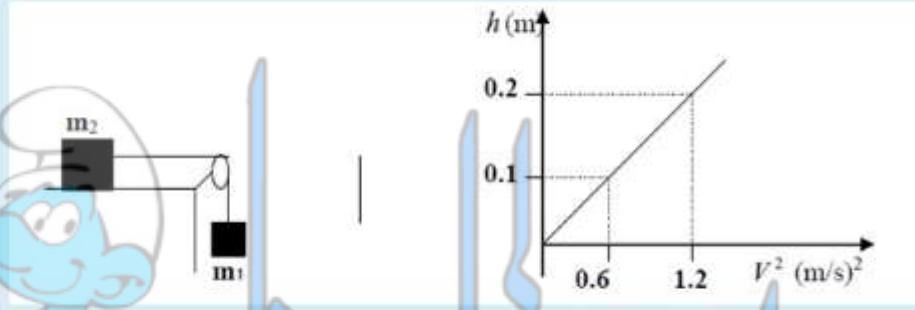
Question 6

Correct

Mark 1.00 out of 1.00

Flag question

In the conservation of energy experiment, a student obtained data plotted in the following figure. If the masses m_1=0.15 kg and m_2=0.2 kg, the acceleration of gravity (in m/s^2) is:



Select one:

- a. Non of above
- b. 11
- c. 7 ✓
- d. 9
- e. 13

The correct answer is: 7

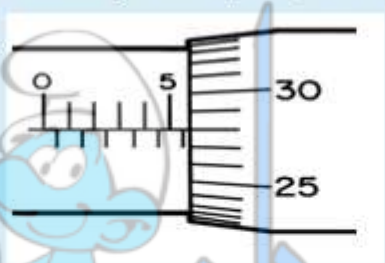
Question 7

Correct

Mark 1.00 out of 1.00

Flag question

In an experiment to measure the density of a small metallic sphere, the diameter of the sphere (d) was measured by micrometer shown in the figure, and the mass of the sphere M was $M \pm \Delta M = (2.1 \pm 0.05)$ g. calculate the density of the sphere, density ± error density



Select one:

- a. 20792 ± 602.97
- b. 10762 ± 302.96
- c. 30552 ± 56.77
- d. Non of above
- e. 24893 ± 652.98

The correct answer is: 20792 ± 602.97

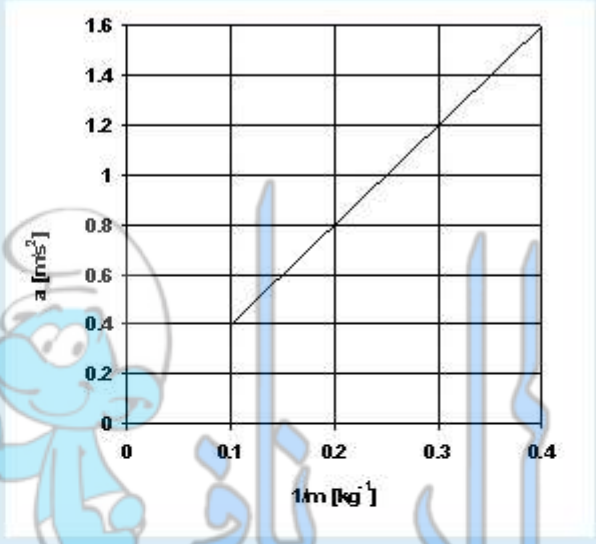
Question 8

Correct

Mark 1.00 out of 1.00

Flag question

In the Newton's second law experiment, the relation between the acceleration (a) and the total mass (m) with a constant force (F) is shown in Figure. The applied force on the system (in N) is:



Select one:

- a. 12 N
- b. 8 N
- c. Non of above
- d. 16 N
- e. 4 N ✓

The correct answer is: 4 N